

Azure Landing Zone & Enterprise Cloud / DevSecOps Architecture

Interview Preparation Notes | Hindi + English

Ye notes 2 diagrams se banaye gaye hain — pehla diagram Azure Landing Zone ke basics cover karta hai, aur doosra diagram poora End-to-End Enterprise Azure Cloud aur DevSecOps Architecture cover karta hai (CAF, Governance, IAM, Networking, Security, DevSecOps pipeline etc).

PART 1 — Azure Landing Zone: Basics

1. What is Azure Landing Zone?

Hindi: Ek pehle se configured, secure, scalable, reliable Azure infrastructure environment hai, jisme best practices ke saath application workload deploy kiya jata hai. Kabhi bhi directly copy-paste nahi karna hota — properly design karke hi banaya jata hai.

English: Azure Landing Zone is a pre-configured, scalable, reliable and secure Azure infrastructure environment where application workloads are deployed following best practices.

2. Azure Landing Zone ke 8 Core Components

- **Hierarchy** — Management Groups / Subscriptions
- **Identity** — Entra ID, Groups, Users, PIM, RBAC
- **Networking** — Hub and Spoke, Flat network, VNet, Subnet, etc.
- **Governance** — Azure Policy, Initiatives, Blueprints
- **Security** — WAF, Firewall, Private Endpoint, NSG etc.
- **Monitoring & Logging** — Azure Monitor, Log Analytics Workspace
- **Backup & DR** — RTO (Recovery Time Objective) aur RPO (Recovery Point Objective)
- **Cost Optimization** — Azure Advisor

3. Management Group Hierarchy in Azure

Sabse top pe ek **Tenant Root Group** hota hai. Uske neeche hierarchy is tarah banti hai:

- **Tenant Root Group** (top level)
 - **Platform Management Group**
 - Identity MG, Connectivity MG, Management MG
 - **Landing Zone Management Group**
 - Workload layer: Dev, Test, UAT subscriptions/resource groups
 - **Decommissioned Management Group**
 - **Sandbox Management Group** (POC, UAT experiments)

Identity MG: Shared Services, Entra ID, Authentication, Authorization, Zero Trust foundation ka base hai.

Connectivity MG: Hub VNet, Firewall, VPN Gateway, Bastion jaise shared networking resources yahin milte hain (Resource Group ke andar). **Management MG:** Monitoring aur Log Analytics Workspace jaise centralized management resources hote hain.

4. Platform Subscription vs Landing Zone Subscription

Platform Subscription	Landing Zone Subscription
Shared Services ke liye use hota hai	Actual application workload ke liye use hota hai
Hub networking, Identity, Management resources	Spoke VNets, Dev/Test/UAT/Prod workloads

5. Hub and Spoke Architecture

Landing zone banate waqt ye recommended network architecture hoti hai. **Central Hub** shared services host karta hai — Hub VNet, Firewall, VPN Gateway etc. **Multiple Spoke VNets** application workload host karte hain aur ye VNet Peering ke through Hub se connect hote hain.

6. Landing Zone banane ki Best Practices

- **Hub VNet (central layer)** — Shared/critical services: Azure Firewall, VPN Gateway, ExpressRoute, DNS, Bastion
- **Spoke VNet (workload layer)** — Har spoke usually ek application ke infra resources rakhta hai jaise VM, VMSS, ACI, App Services etc. Spoke workloads ko direct internet exposure nahi diya jata.
- Spoke-to-Spoke communication kabhi directly nahi hota — hamesha Hub Firewall se hote hue UDR (User Defined Route) ke through jata hai.
- PaaS services ko spoke network me hamesha **Private Endpoints** ke through access karo.
- Jahan possible ho, **multi-region design** ke saath architecture deploy karo (High Availability ke liye).

7. Hub and Spoke ke Advantages

- **Cost Optimization** — shared services se duplication kam hoti hai
- **Single Firewall** — saare workloads ke liye ek hi centralized firewall
- **Centralized Security**
- **Scalability**
- **Isolation** — har workload apne spoke me isolated rehta hai

PART 2 — End-to-End Enterprise Azure Cloud & DevSecOps Architecture

Goal: Governance, Hierarchy, IAM, Networking, Compute, Backup, DR, Security, Cost Optimization, Best Practices, Monitoring, High Availability, Scalability aur Reliability ko cover karte hue ek complete enterprise-grade Azure architecture design karna.

1. CAF — Cloud Adoption Framework

CAF Microsoft ka recommended method hai successful cloud adoption ke liye. Iske phases:

- **Strategy** — Business justification aur expected outcomes define karna (learn.microsoft.com/azure/cloud-adoption-framework/strategy)
- **Plan** — Organization ko cloud ke liye prepare karna (learn.microsoft.com/azure/cloud-adoption-framework/plan/prepare-organization-for-cloud)
- **Ready (Landing Zone)** — Secure, governed aur scalable Azure foundation provide karta hai
- **Adopt** — Is phase me workload deploy aur migrate kiya jata hai
- **Govern** — Compliance aur policy enforcement
- **Manage** — Deployment ke baad ki operations

2. Cloud Adoption ke Key Questions

- Organization Azure pe kyun move kar raha hai?
- Cloud se kya business value milega?
- Success measure karne ke liye kaunse KPIs use honge?
- Current environment assess karke migration roadmap banana

3. 6 R's of Migration (Migration Strategies)

Migration approach decide karne ke liye industry-standard framework — Rehost, Replatform, Repurchase, Refactor/Re-architect, Retire, aur Retain. Har workload ke liye sahi 'R' choose karke migration plan banaya jata hai.

4. Migration Plan

- **Migrate** — Lift and Shift, Azure Migrate tool, Database Migration Service, VMware Migration
- **Modernization** — Monolithic applications ko Microservices me convert karke AKS, ACI, Serverless/Functions, App Services pe deploy karna

5. Hierarchy

Management Groups aur Subscriptions ke through Governance, Compliance, Billing aur IAM/RBAC manage kiya jata hai.

6. Identity

- **DevOps Group** — Contributor access
- **Developer Group** — Reader access
- Entra ID ke core components: Groups, Users, RBAC, PIM (Privileged Identity Management), Managed Identity, Conditional Access

7. Governance

- Azure Policy
- Tagging aur Naming Standards
- Budget aur Cost Management
- Resource Locks

8. Networking

- Hub and Spoke ya Flat Network topology
- Hub me kya hota hai: shared/critical services
- Spoke me kya hota hai: application workload
- VPN Gateway, Azure Firewall, NSG, WAF
- Private Endpoints, ExpressRoute, DDoS Protection

9. Management (Backup, DR, Monitoring)

- Azure Monitor
- Log Analytics Workspace
- Diagnostic Logs

10. Security

- Firewall, NSG (Network Security Group), ASG (Application Security Group)
- WAF (Web Application Firewall)
- Microsoft Sentinel (SIEM)
- Microsoft Defender for Cloud

11. Cost / Budget Optimization

- Reservations
- Savings Plans
- Spot Instances

12. Automation, Observability & Ops

- Security, Tagging aur Policy compliance **Terraform Automation** se enforce karna
- **Observability** — Incident Management, ITSM, Alerting
- Backup and DR planning

Reference Architecture Example (Sample)

Ek sample enterprise setup jisme user traffic Azure Front Door aur CDN se hoke aata hai, phir application ke Virtual Networks tak jaata hai — har VNet me Frontend Subnet aur Backend Subnet hota hai (multi-region setup).

- Users → Azure Front Door → CDN
- Traffic route hota hai VNet(s) tak — har region me Frontend Subnet + Backend Subnet
- Yeh pattern High Availability aur global traffic distribution ke liye use hota hai

PART 3 — DevSecOps

1. What is DevSecOps?

DevSecOps, DevOps pipeline ke har stage me Security ko integrate karne ka practice hai — security ko end me ek gate banane ke bajaye, code-to-deployment ke har step me shift-left approach se check karna.

2. Deployment se pehle kaunse Security Checks chalane chahiye?

In tools se code aur infrastructure ka security scan kiya jata hai:

- Snyk
- Checkov
- tfsec
- Gitleaks
- TruffleHog
- Trivy
- BlackDuck
- Prisma Cloud
- Chef InSpec
- TFLint
- SuperLinter
- SonarQube
- Terracost
- Terrascan
- VSTest

3. Terraform ko kaise secure karte hain?

- **State Locking** — concurrent state modification avoid karne ke liye
- **Least Privilege** — service connections/identities ko minimum required access dena
- **Child-Parent Modules** — reusable aur controlled module structure
- **Pull Requests** — peer review ke bina koi bhi change directly apply nahi hota

4. Terraform me Secrets kaise secure karte hain?

Secrets ko **Azure Key Vault** me store karke Terraform se reference kiya jata hai — hardcoded values kabhi use nahi karte.

5. DevSecOps Tools (used in project)

Terraform, Git, Pipelines (Azure DevOps/GitHub Actions), tfint, tfsec, TruffleHog, Gitleaks, Checkov, Chef InSpec.

6. Zero Trust Security in Azure

Zero Trust ka principle hai — 'Never trust, always verify'. Har request ko explicitly verify kiya jata hai (identity, device, location), least-privilege access diya jata hai, aur assume kiya jata hai ki breach kabhi bhi ho sakta hai — isliye continuous monitoring aur segmentation zaroori hai.

7. Azure Landing Zone ko kaise secure karte hain?

- Network isolation via Hub-Spoke + NSG/ASG
- Identity: RBAC, PIM, Conditional Access
- WAF + Firewall + Private Endpoints
- Sentinel + Defender for continuous threat detection

8. Landing Zone ka Compliance & Governance kaise manage karte hain?

Microsoft Sentinel, Azure Policy, Initiatives aur Blueprints ke through continuous compliance monitor aur enforce kiya jata hai.

9. Complete DevSecOps Pipeline Flow

- Repo clone karna
- Feature branch create karna
- Code push karna
- Pipeline me **terraform plan** chalana aur security scan reports check karna
- Pull Request (PR) raise karna
- Code review milna
- PR ko main branch me merge karna
- Main branch se pipeline trigger hota hai — plan → scan → **apply**

10. Architect ki Primary Responsibilities

- Hands-on implementation
- Screen share karke draw.io pe design teach karna / explain karna
- Mock interviews conduct karna
- Continuous learning — market me improve karte rehna, resume banate rehna

Note: Ye notes dono uploaded diagrams (Azure Landing Zone basics + End-to-End Enterprise DevSecOps Architecture) ke content ko organize karke banaye gaye hain, taaki interview prep aasan ho jaaye.

PART 4 — Quick Revision: Question & Answer (Interview Style)

Sabhi topics ke crisp Q&A ek jagah — last-minute revision ke liye.

Q1. Azure Landing Zone kya hai?

A. Ek pre-configured, secure, scalable, reliable Azure infra environment jisme best practices ke saath application workload deploy hota hai.

Q2. Azure Landing Zone ke main components kya hain?

A. Hierarchy, Identity, Networking, Governance, Security, Monitoring & Logging, Backup & DR, Cost Optimization.

Q3. Management Group Hierarchy explain karo.

A. Tenant Root Group ke neeche Platform MG (Identity, Connectivity, Management MG) aur Landing Zone MG (Dev/Test/UAT workloads), plus Decommissioned aur Sandbox MG.

Q4. Platform aur Landing Zone subscription me kya farak hai?

A. Platform subscription shared services ke liye hoti hai; Landing Zone subscription actual application workload ke liye hoti hai.

Q5. Hub and Spoke Architecture kya hai?

A. Recommended network architecture jisme central Hub shared services (Firewall, VPN GW) host karta hai aur multiple Spoke VNets application workload host karke VNet Peering se Hub se connect hote hain.

Q6. Landing Zone banane ki best practices kya hain?

A. Hub me critical services (Firewall, VPN GW, ExpressRoute, DNS, Bastion); Spoke me app resources; spoke-to-spoke traffic hamesha firewall/UDR se; PaaS access Private Endpoints se; multi-region design jahan possible ho.

Q7. Hub and Spoke ke advantages kya hain?

A. Cost optimization, single centralized firewall, centralized security, scalability, isolation.

Q8. CAF (Cloud Adoption Framework) kya hai?

A. Microsoft ka recommended framework hai successful cloud adoption ke liye, jisme phases hain: Strategy, Plan, Ready (Landing Zone), Adopt, Govern, Manage.

Q9. Organizations Azure pe kyun move karte hain?

A. Business value, cost efficiency, scalability aur measurable KPIs ke liye — current environment assess karke migration roadmap banate hain.

Q10. 6 R's of Migration kya hain?

A. Rehost, Replatform, Repurchase, Refactor/Re-architect, Retire, Retain — har workload ke liye best migration strategy choose karne ka framework.

Q11. Migration Plan me kya steps hote hain?

A. Migrate (Lift and Shift, Azure Migrate, DMS, VMware Migration) aur Modernization (Monolithic → Microservices → AKS/ACI/Serverless/App Services).

Q12. Identity aur IAM me kya cover hota hai?

A. DevOps Group (Contributor), Developer Group (Reader), Entra ID ke Groups/Users/RBAC/PIM/Managed Identity/Conditional Access.

Q13. Governance me kya aata hai?

A. Azure Policy, Tagging & Naming standards, Budget & Cost Management, Resource Locks.

Q14. Networking me kya components hote hain?

A. Hub and Spoke/Flat network, VPN Gateway, Azure Firewall, NSG, WAF, Private Endpoints, ExpressRoute, DDoS Protection.

Q15. Security me kya-kya tools/concepts aate hain?

A. Firewall, NSG, ASG, WAF, Microsoft Sentinel, Microsoft Defender for Cloud.

Q16. Cost/Budget optimization kaise karte hain?

A. Reservations, Savings Plans, Spot Instances ke through.

Q17. DevSecOps kya hai?

A. DevOps pipeline ke har stage me security ko integrate karna (shift-left approach) — end me ek gate banane ke bajaye.

Q18. Deployment se pehle kaunse security checks chalane chahiye?

A. Snyk, Checkov, tfsec, Gitleaks, TruffleHog, Trivy, BlackDuck, Prisma, Chef InSpec, TFLint, SuperLinter, SonarQube, Terracost, Terrascan, VSTest.

Q19. Terraform ko kaise secure karte hain?

A. State Locking, Least Privilege access, Child-Parent module structure, mandatory Pull Requests.

Q20. Terraform me secrets kaise secure karte hain?

A. Azure Key Vault me store karke reference karte hain — hardcode kabhi nahi karte.

Q21. Zero Trust Security explain karo.

A. 'Never trust, always verify' principle — har request explicitly verify hoti hai, least-privilege access diya jata hai, aur continuous monitoring ke saath breach hamesha assume kiya jata hai.

Q22. Azure Landing Zone ko secure kaise karte hain?

A. Hub-Spoke network isolation + NSG/ASG, RBAC/PIM/Conditional Access, WAF+Firewall+Private Endpoints, Sentinel+Defender monitoring.

Q23. Compliance aur Governance kaise manage karte hain?

A. Microsoft Sentinel, Azure Policy, Initiatives aur Blueprints ke through continuous monitoring aur enforcement.

Q24. Complete DevSecOps pipeline flow batao.

A. Clone → feature branch → push code → pipeline me plan + scan reports check → PR raise → review → merge to main → main se pipeline run (plan → scan → apply).

Q25. Architect ki primary responsibilities kya hoti hain?

A. Hands-on implementation, draw.io pe design explain karna, mock interviews lena, continuous learning aur improvement karte rehna.